Supplementary Notes for Granularity Ablation

1 Experimental Setup

The granularity ablation isolates how the divisibility of a resource block affects useful resource utilization under budget-constrained buyers. A resource block costs 60 and is split into 3, 30, or 300 indivisible tokens, so that the minimum tradable unit is $1/N$ of a block and the minimum ticket is $60/N$. All seller capacity is escrowed into a constant-product pool. We vary the number of buyers from 100 to 300, each with a fixed budget between 60 and 120, and let buyers trade whole tokens in a paired random order. Across all conditions, sellers, physical supply, buyer cash, and the pricing rule are held fixed; only the seller inventory recorded on-chain can be delivered. The primary metric is useful resource utilization.

2 Results and Analysis

As shown in `ablation_allocation.pdf`, two findings emerge. First, finer lots produce higher useful utilization at every buyer count: the ordering $3 < 30 < 300$ is preserved throughout the range of 100–300 buyers. This gap is not a level shift confined to a single demand regime; it persists as demand grows, indicating that granularity affects the allocation outcome rather than merely shifting its baseline.

Second, the underlying mechanism is an entry-barrier effect. Splitting a block into N tokens lowers the minimum ticket from the cost of a whole block to $60/N$. Budget-constrained buyers who cannot afford a whole block, or who can afford only a small fraction of one, are excluded when $N = 3$ but can participate when $N = 300$. Finer granularity also allows a buyer to purchase closer to actual demand and reduces overpurchase of the last indivisible token. If the participant distribution is skewed toward small budgets, coarse tokenization silently caps useful utilization even when physical supply is adequate. Finer tokenization lowers the entry barrier and admits more participants, so that a larger share of capacity is matched to demand rather than stranded in unsold remainders.

This static entry-barrier effect is reinforced by the endogenous price path of the AMM. A buyer may initially be able to afford the minimum ticket at the pre-trade quote, but as resource inventory is purchased, the constant-product quote rises. Under coarse granularity, each trade must involve at least 1/3 of a block, so a buyer must commit a larger absolute amount and exerts a larger price impact per transaction; under fine granularity, the minimum trade is only 1/300 of a block, allowing smaller purchases and limiting the quote increase faced by subsequent buyers. Consequently, even buyers whose budgets were sufficient at the initial price can differ in subsequent affordability: coarse tokenization can price them out after earlier purchases, whereas fine tokenization lets them continue to participate in smaller increments.